Ten venues, three days

We can deliver this by end of November, with about ten days of float. The scope below is what I think we should commit to. Three things need a client decision this week, and one thing we found in our own code has to be fixed before a single unit is built.

WHAT WE'D BE COMMITTING TO		
WEARABLES **5,000–10,000** Existing radio module in a custom case. No new PCB.	VENUES **10** One gateway each — venue-level triggers, not in-venue zones.	DURATION **3 days** Devices run the whole event without recharging.
THE EXPERIENCE **Arrive → ping** Guest registers on the web app; a message reaches their phone on arriving at a venue.	DELIVERY **30 Nov** All ten venues live. ~10 days float in the plan.	DATA **Name + contact** Held in the EU, deleted 30 days after the event.

00 Can we do it

Short answer: yes, and this is the first version of the brief where that's true.

The client opened at 50,000 units across fifteen properties. At that size it was not deliverable — the radio physics didn't work, and neither did the manufacturing calendar. At **5,000–10,000 units, ten venues and one gateway apiece**, almost every hard constraint disappears:

WHAT THE SMALLER SCOPE REMOVES	
CONSTRAINT AT 50K	AT 5–10K
Custom PCB design, tooling, FCC campaign	`Existing module in a custom case — no new board, no new certification campaign` GONE
Sea freight leaving China by 5 October	`Volume small enough to air-freight, or build without export at all` GONE
Bulk lithium battery shipping restrictions	`Routine volume` GONE

CONSTRAINT AT 50K	AT 5–10K
SMS short code — 8–12 week provisioning	~50–100k messages fits standard carrier registration GONE
~180 gateways, 15 installs, union labour programme	~20 gateways, 10 small installs MANAGEABLE
Radio channel capacity 50× oversubscribed	Solvable — but see §04, it needs a firmware change WATCH
Device identity collisions across the fleet	Still real. Must be fixed before build FIX FIRST
US texting law exposure	Unchanged by scale UNCHANGED

What's left is a project with two known problems, both of which we can fix ourselves, and a legal exposure that needs a lawyer rather than an engineer. That's a normal project.

01 In and out of scope

Worth agreeing this explicitly before the client fills the gaps with their own assumptions.

✓ WHAT THEY GET

- Guest registers once on a web app and is bound to a device
- A message to their phone when they arrive at any of the ten venues
- Their journey across venues over three days, as data
- A live operations view of all ten venues and every gateway
- Aggregate reporting: venue-to-venue flows, dwell, cohort behaviour
- Devices that run the full three days without recharging

✗ WHAT THEY DON'T — SAY SO NOW

- **In-venue zone granularity.** One gateway per venue means "arrived at the Bellagio", not "near the east bar". Zones inside a venue are more gateways and more money.
- **Guaranteed device recovery.** Ten venues means no single exit. Assume we don't get them back unless they fund a scheme.
- **Synchronised effects across venues.** No shared radio domain between properties — cross-venue timing is network-accurate, not show-accurate.
- **Messages at any hour.** US law caps texting at 8am–9pm local. A late-night moment can't be a text.
- **Precise location.** We know they passed a gateway. We don't track them continuously, and we shouldn't offer to.

The last two are worth leading with rather than burying. A client who hears "we deliberately don't track people continuously, and here's why" tends to trust the rest of the pitch more, not less.

02 Three things

Ranked by what they'd cost us if they went wrong.

1 · LEGAL

US texting law

Sending marketing texts to US mobiles without watertight written consent carries statutory damages of **$500–$1,500 per message**. At 50,000–100,000 messages that is the only number in this document that could end the company. It does not get smaller with scale.

Mitigation: US counsel drafts the consent wording before registration opens. Push notifications instead of SMS wherever possible — no exposure, no cost. Budget for the legal review.

2 · DEFECT

Device identity collisions

Our device addresses are derived from the chip's serial number into a space sized for a hundred bands. At 10,000 units roughly **1,500 guests would share an identity with someone else** — receiving each other's messages, journeys merged. §03.

Mitigation: one week of firmware plus a provisioning tool we half-planned already. Must land before any unit is programmed.

3 · AVAILABILITY

One gateway per venue

A single gateway failure takes a whole venue dark for the rest of a three-day event, and we can't drive a spare to ten properties fast enough. It also concentrates all of a venue's radio traffic onto one channel. §04.

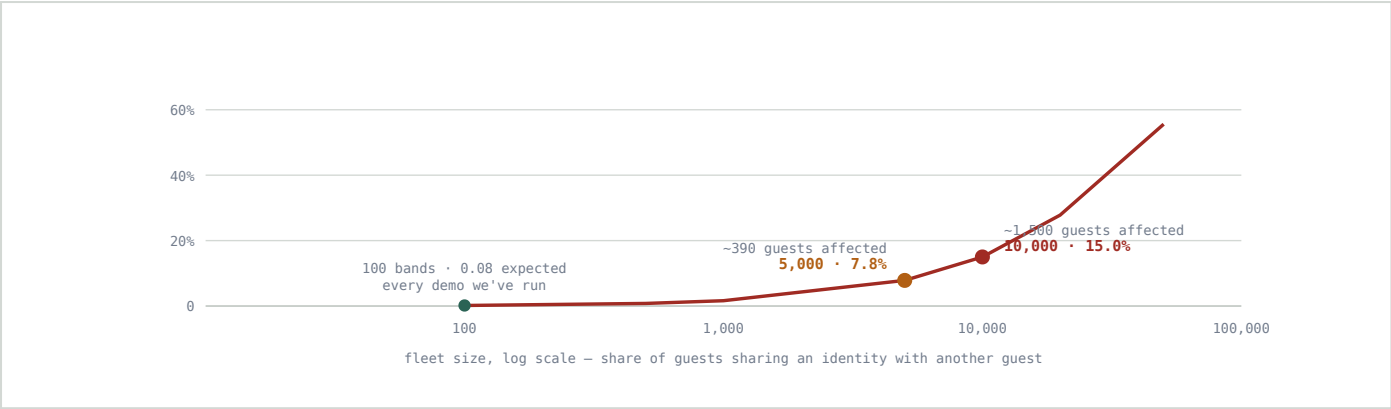
Mitigation: two gateways per venue. Twenty units instead of ten — the cheapest insurance in the plan by a wide margin.

03 The address defect

Found while modelling the fleet at scale. It would have shipped silently.

Every band derives its identity from the chip's factory serial number, hashed into a 16-bit address. There are 61,438 addresses available. The comment in our own header is honest about the assumption — it says collisions are unlikely for a hundred bands and can be resolved later by a registration flow. **At 5,000–10,000 units, "later" has arrived.**

Collisions scale with the square of the fleet. Fifty times the devices is two and a half thousand times the collisions:



Flat where we tested, vertical where we're going. The canonical shape of a defect that passes every demo.

The reason it matters more than a dropped radio frame is that this one number is simultaneously the radio address, the registration key and the analytics key. Two guests sharing it are **the same person** to the whole system: they get each other's messages, their journeys merge, their registration records cross. At an event where we hold names and phone numbers, that's a guest-experience failure, a data-protection incident and — if a message lands on the wrong phone — a consent problem on the statute in §02.

> **THE FIX IS A WEEK, AND WE'D HALF-PLANNED IT ALREADY**
>
> At provisioning we touch every device anyway to attach its label. We read the derived address over the dock, check it against a registry, and where it's taken we allocate a free one and write it to the device. Around 92% of units need nothing. There are two work items — a small firmware change and the provisioning tool — and the empty `labels/` and `registration/` folders in the repo suggest this was already on someone's list. It just has to land *before* we program anything.

> **WORTH SAYING TO THE CLIENT**
>
> This is a good story, not an embarrassing one. We modelled their fleet at full scale in August and found a defect that would have put messages on the wrong guests' phones in November. That's the difference between a hardware partner and a beacon reseller, and it's worth more to them than the discount they were expecting to negotiate.

04 Why 20 gateways, not 10

Two independent reasons, and the second one is counter-intuitive.

REASON ONE — NO REDUNDANCY

With one gateway per venue, any single failure — a power blip, a dead SD card, a cellular outage, a cleaner unplugging something — takes that venue completely dark for the remainder of a three-day event. We would find out from the client. Across ten properties in a city, we cannot get an engineer to a dead gateway quickly enough to matter. A second unit per venue is straightforward, cheap at this quantity, and turns a total venue outage into a degraded one.

REASON TWO — FEWER GATEWAYS MAKES THE RADIO PROBLEM WORSE

This is the part that surprised me. Radio contention is per-channel, and devices have to sit on the channel their gateway is using. With five gateways in a venue, that venue's devices can be spread across five channels. **With one gateway, every device at that venue is on one channel.** Cutting the gateway count concentrated the load rather than reducing it.

RADIO LOAD AT A BUSY VENUE — ~2,000 DEVICES IN EARSHOT

DESIGN	FRAMES/SEC	CHANNEL CAPACITY	RESULT
Free-running beacons, one gateway	1,000	~500	OVER
Free-running, two gateways, two channels	500	~500	AT THE LIMIT
Gateway-triggered transmit	10–30	~500	20× MARGIN

The proper fix is a firmware change: instead of devices beaconing constantly, the gateway broadcasts and a device only answers when it hears one, then goes quiet for that venue for a while. Radio load then scales with *people walking past* rather than with how many devices exist. It also happens to be what makes the three-day battery work, because a device that only transmits on cue is a device with its radio off the rest of the time.

> **THE COMMERCIAL VERSION OF THIS**
>
> The firmware fix is the right answer and it's in the plan. The second gateway per venue is *insurance against that firmware not landing on time* — with two channels in play, free-running beacons just about work as a fallback. So twenty gateways buys us redundancy and a plan B on the single piece of work with the most schedule uncertainty. If the client pushes back on anything, this should not be it.

05 Cost drivers

Not a quote — the shape of one, and which levers actually move the number.

WHERE THE MONEY IS

LINE	DRIVER	SENSITIVITY
Radio modules	5,000 vs 10,000 units	Largest single line and near-linear with unit count. This is *the* number to get quoted this week — everything else is noise beside it.
Battery cells	Per unit, plus DG paperwork	Small per unit. Fit a generous cell — see below.
Case tooling	One-off	Our last quote was $2,130 for a simple part. A two-part case for module and battery is likely $4–6k. Needs a fresh quote against the actual geometry, but it is not where the money goes.
Case parts	Per unit	Pennies. At 5,000 the tool amortises to well under a dollar a unit.
Assembly	5,000–10,000 hand builds	Underrated. Module, cell, case, label, test, charge — a few minutes each is 6–12 person-weeks. Decide now whether we do it or a contract assembler does.

LINE	DRIVER	SENSITIVITY
Gateways	20 units	Pi, custom HAT, PoE, enclosure, cellular router per venue. Small money at this quantity.
Connectivity	10 SIMs, one month	Rounding error.
Messaging	50–100k messages	Roughly $550–$1,100 on SMS. Push notifications are free — worth designing the registration flow to earn the install.
Install + on-site	10 venues, 3 event days	Travel, crew, house labour where properties require it, and staffing for three days. Confirm the labour position property by property before pricing.
Legal	Fixed	US counsel on consent wording, plus a data protection impact assessment. Both non-optional. Small next to the exposure they cover.

TWO COMMERCIAL JUDGEMENTS WORTH MAKING DELIBERATELY

- **Module-in-a-case is the right call at this size, but know why.** We're trading unit cost for schedule and NRE. A custom board would be cheaper per unit but adds design, prototyping and certification we do not have time for. The crossover is somewhere north of 10,000 units — so if this becomes a repeat programme, the custom board pays for itself on run two. Worth framing to the client as exactly that: this run proves it, the next one industrialises it.
- **Fit a bigger battery than the maths needs.** Our current draw figure is an estimate, not a measurement, and it's the input to a three-day runtime commitment. The case is custom, so there's room. Going from a 150 mAh cell to 300 mAh costs almost nothing per unit and removes an entire category of on-site failure. We should still do the firmware gating — but we shouldn't bet the event on an unmeasured number when margin is this cheap.

06 Client decisions

These gate our work. Each one has a date after which it costs us float.

WHAT WE NEED, AND WHEN

DECISION	NEEDED BY	WHAT IT BLOCKS
5,000 or 10,000?	1 Sep	Module order. Longest lead item and the biggest cost line. We cannot order to a range.
Which ten venues	5 Sep	Site surveys, gateway siting, backhaul, labour position. Nothing in the deploy track starts without it.
Are any of them casinos?	This week	Gaming licensees run vendor security reviews that are slower than any venue IT process. If the answer is yes, that clock starts now.
What the message actually is	15 Sep	Whether it's an offer, wayfinding or a game mechanic decides how it's classified under US texting law — the single largest lever on our legal exposure.
Do guests keep the device?	15 Sep	Case design, unit economics, and whether we need collection points and staff.
Who is the data controller?	15 Sep	We want it to be them, in writing, before anyone collects a name. It substantially reduces our obligations and our liability.

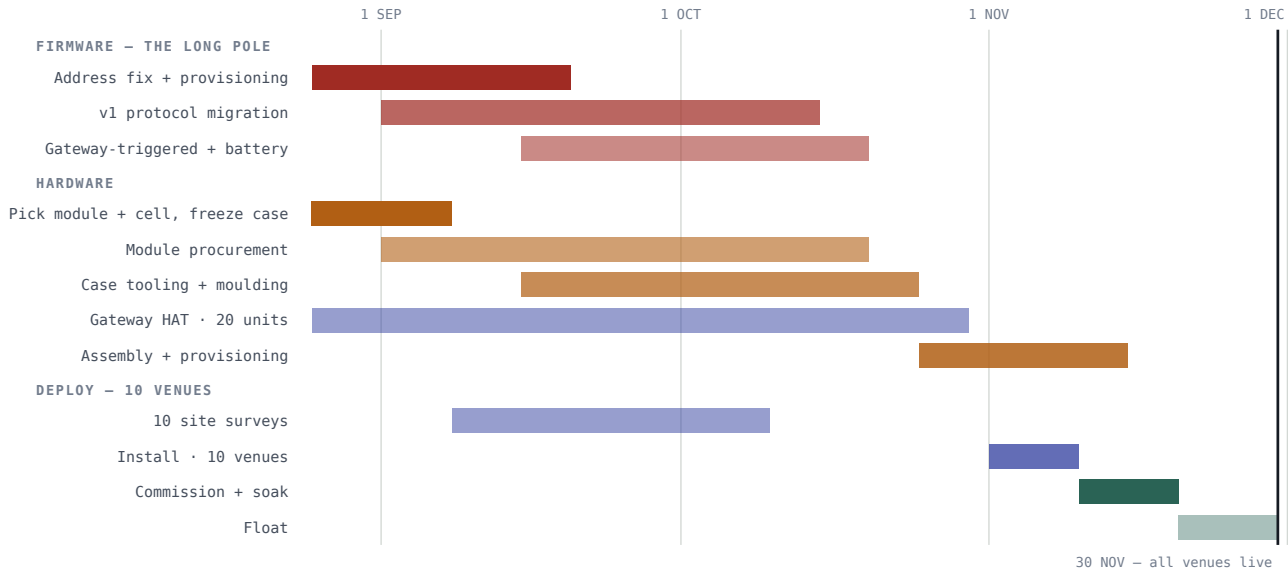
DECISION	NEEDED BY	WHAT IT BLOCKS
Sign-off on scope	8 Sep	Everything. The §01 exclusions especially — those need to be agreed, not assumed.

DO NOT SIGN WITHOUT THIS ONE

The data controller question isn't paperwork. If the client determines what the data is collected for, they carry the bulk of the regulatory obligation and we act on their instructions. If we determine it — including by using the journey data for our own analytics afterwards — we take on their obligations too. That second position is worth real money and real risk, and it's decided by a contract clause, not by engineering. If we want to keep anything from this event for our own product, it has to be aggregate only and it has to be written into the agreement up front.

07 The plan

Three tracks. For the first time, hardware isn't the long pole — firmware is.



Everything lands by 20 November. The float sits behind the firmware track, which is where the unknowns are — we shouldn't spend it early on hardware.

Two things to note when you're talking to the client about dates. **The firmware track starts now and finishes in October** — it has no external dependencies, which is why it's the one we can actually control. And **the hardware track is entirely gated on the unit-count decision**: module procurement is the longest bar on the chart, and it cannot start until someone says 5,000 or 10,000. If that decision slips past the first week of September, the float goes with it.

08 This week

What I'm doing, and where I need you.

01 **Fix the device address derivation**

Firmware override plus the provisioning registry. Has to land before we program a single unit. I'll start this.

02 **Quote the modules at 5,000 and 10,000**

Biggest cost line, longest lead item, and it gates the whole hardware track. We need real numbers to put a price in front of the client at all.

03 **Get the venue list — and ask whether any are casinos**

Everything in the deploy track is downstream of this, and if there's a gaming licensee in the list their vendor review clock needs to start now, not in October.

04 **Brief US counsel on the texting consent wording**

Highest-severity item in the document and the cheapest to close. Needs the message classification from the client to be useful, so it's worth pairing the two conversations.

05 **Measure the actual current draw**

A bench afternoon. It's the input to a three-day runtime commitment we're about to make contractually.

06 **Get scope sign-off, exclusions included**

§01. The exclusions matter more than the inclusions — if in-venue zones or device recovery get assumed into the deal now, they get discovered in November.

07 **Price it at 20 gateways, not 10**

Redundancy plus a fallback if the firmware slips. It's the item most likely to get value-engineered out, and the one we should defend hardest.

08 **Decide who assembles 5,000–10,000 units**

6–12 person-weeks of work sitting quietly in the middle of November. Us or a contract assembler — but decided now, not then.

ÆTHERFIELD · Las Vegas scope and delivery plan · prepared for Shaun · 24 August 2026
Collision figures computed against the band address range 0x1000–0xFFFD (61,438 addresses) assuming uniform hash distribution. Radio, battery and schedule figures are modelled from the assumptions stated in each section and move with them. Cost lines are drivers, not quotations. Regulatory summaries are orientation for a conversation with counsel, not legal advice.